Operating Room Scheduling Optimization

1. Introduction

Efficient scheduling of surgeries is a critical challenge in hospitals. Poor planning can lead to delays, underutilized operating rooms, and increased waiting time for patients. This project focuses on optimizing the allocation of surgeries to available operating rooms using data-driven techniques.

The aim is to create a schedule that minimizes total time while ensuring that no two surgeries overlap in the same room.

2. Objective

The primary objectives of this project are:

- To assign surgeries to available operating rooms efficiently
- To avoid overlapping schedules
- To minimize the overall completion time of surgeries
- To improve utilization of hospital resources

3. Methodology

This project uses Constraint Programming with the help of Google OR-Tools to solve the scheduling problem.

Key steps followed:

1. Data Preparation

- Each surgery has a predicted duration
- Available operating rooms are defined

2. Model Creation

- A CP-SAT model is built using OR-Tools
- Decision variables include:
- Start time of each surgery
- End time of each surgery
- Room assignment

3. Constraints Applied

- No overlapping surgeries in the same room
- Each surgery must be assigned to one room
- Start and end times must follow duration rules

4. Objective Function

- Minimize the total completion time of all surgeries

4. Tools and Technologies Used

- Python – Core programming
- OR-Tools – Optimization solver
- Pandas – Data handling
- Power BI – Dashboard and visualization
- Matplotlib – Basic visualization

5. Results

The model successfully generated an optimized schedule for all surgeries.

- Total surgeries scheduled: 30
- Average duration: ~147 minutes
- Surgeries were distributed across 3 operating rooms (OR1, OR2, OR3)
- No overlapping schedules were observed

The output includes:

- Surgery ID
- Assigned room
- Start time
- End time

6. Dashboard Insights

A Power BI dashboard was created to analyze and present the results.

Key observations:

- Room Utilization

OR2 handled the highest total duration of surgeries, indicating heavier usage.

- Balanced Distribution

Surgeries were fairly distributed across all rooms, avoiding overloading a single room.

- Time Efficiency

The optimized schedule ensures minimal idle time between surgeries.

- Clear Timeline View

A Gantt chart provides a visual representation of the entire schedule, making it easy to understand sequencing and allocation.

7. Conclusion

This project demonstrates how optimization techniques can significantly improve operational efficiency in a healthcare setting. By using constraint programming, we were able to create a structured and conflict-free surgery schedule.

The solution not only ensures better utilization of operating rooms but also helps in reducing delays and improving overall workflow.

8. Future Improvements

- Incorporate emergency surgery handling
- Add surgeon and staff availability constraints
- Include real-time scheduling adjustments
- Integrate machine learning for duration prediction

9. Summary

This project highlights the practical application of data science and optimization in solving real-world problems. The combination of Python, OR-Tools, and Power BI provides both a strong backend solution and an intuitive front-end visualization.